ISOENERGETIC KAM, CONFINEMENT AND THE RESTRICTED THREE-BODY PROBLEM: BLUEPRINT

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Abstract. A proof plan for the four frozen statements of track I: the isoenergetic a-posteriori KAM theorem (I-G1), all-time action confinement for two degrees of freedom (I-HEAD), the Kepler bordered determinant (I-KEPLER), and the near-circular region of the restricted circular planar three-body problem (I-R3BP). Results of the blueprint of track K are cited by their labels [K:`bp:p1:...`] and [K:`bp:p2:...`]. Every result is labelled and cites its pinned source.

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the four frozen theorems of track I and the frozen definitions they use: the fifteen definitions of `KAM/Frozen/Iso/Defs/` (owner approval D20, 27 October 2026) and the definitions of `KAM/Frozen/Defs/` of track K. The Lean declarations are authoritative; the statements below are faithful transcriptions, with the quantifier order and the order in which constants are chosen exactly as in Lean. Results of track K's blueprint are cited as [K:`bp:p1:...`] and [K:`bp:p2:...`]; they are not re-proved here.

1.1. Conventions.

- Ambient spaces are $\mathbb{R}^m = (\texttt{Fin}\ m \to \mathbb{R})$ and $\mathbb{C}^m = (\texttt{Fin}\ m \to \mathbb{C})$. Angles are $\theta \in \mathbb{C}^n$, actions $I \in \mathbb{C}^n$, phase points $z = (\theta, I) \in \mathbb{C}^n \times \mathbb{C}^n$. Functions of θ are 1-periodic, as in Part 0 of track K ([K:`bp:p0:conventions`]).
- *Norms.* $|v| = \|v\| = \max_i |v_i|$ on $\mathbb{R}^m$ and $\mathbb{C}^m$; on a product the maximum of the two component norms. This is the norm of `Metric.ball`, of `dist` in `actionNbhd`, of $\|DH\|$ (operator norm of a $\mathbb{C}$-linear map, `fderiv`), and of $\|D^2H\|$ (norm of the bilinear map `iteratedFDeriv`). A row-sum bound $\sum_j |M_{ij}| \le c$ for every i is written $|M| \le c$. The norm of a complex number is its modulus.

- The strip $T_\rho = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta_i| < \rho \ \forall i\}$ is open; it contains $\mathbb{R}^n$ and is invariant under $\mathbb{Z}^n$ and $\mathbb{R}^n$ translations. The set `Set.pi univ (Icc 0 1)` $= [0,1]^n$ carries Lebesgue measure, of total mass one; $\langle u\rangle = \int_{[0,1]^n} u(\theta)\,d\theta$ for a function evaluated at real angles. Averages of vectors or matrices are entrywise.
- A map is *holomorphic on an open set* U if it is `DifferentiableOn` $\mathbb{C}$ there. On an open set all derivatives `fderiv`, `iteratedFDeriv` used below are genuine (Lean totality, Remark 1.13).
- Real and complex points. A real vector $x \in \mathbb{R}^m$ is identified with $(x_i) \in \mathbb{C}^m$; "$(I : \mathbb{C}) \in S$" for $S \subset \mathbb{C}^n$ means this identification lies in S. The real phase points of $D \subset \mathbb{C}^n \times \mathbb{C}^n$ are the $(\theta, I) \in \mathbb{R}^n \times \mathbb{R}^n$ whose image lies in D.
- *Two phase-space conventions.* In I-G1 and I-HEAD, a phase point is (θ, I) (angles first) and $X_H = (\partial_I H, -\partial_\theta H)$. In the restricted three-body problem a phase point is $z = (q, p)$ with position $q \in \mathbb{R}^2$ and canonical momentum $p \in \mathbb{R}^2$ (position first, as in the Lean carrier $(\texttt{Fin}\ 2 \to \mathbb{R}) \times (\texttt{Fin}\ 2 \to \mathbb{R})$); there `radius` is the Euclidean length, not the max norm. The norm $\|\cdot\|$ of I-HEAD is the max norm; the bounds in the conclusion of I-R3BP use rad.

1.2. **Frozen definitions of track K used by track I.**

Definition 1.1 (Strip, action neighbourhood)**.** $T_\rho = \{\theta \in \mathbb{C}^n : \forall i,\ |\operatorname{Im}\theta_i| < \rho\}$ (`complexStrip` n ρ). For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ \|I - J\| < r\}$ (`actionNbhd`); it is open. Thus for real I: $I \in B_r$ iff $\exists J \in B$ with $\|I - J\| < r$.
Frozen Lean declaration: `KAM.complexStrip, KAM.actionNbhd`
Source: HL19 §2.1; see [K:`bp:p0:domains`].

Definition 1.2 (Diophantine vectors)**.** $\omega \in \mathbb{R}^n$ is (γ, τ)-Diophantine if for every $k \in \mathbb{Z}^n \setminus \{0\}$ $\gamma/\big(\sum_i |k_i|\big)^\tau \le |k \cdot \omega|$ (real power).
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; see [K:`bp:p0:diophantine`].

Definition 1.3 (Hamiltonian vector fields)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and a real phase point z: $X_H(z) = \big(i \mapsto DH(z)(0, e_i),\ i \mapsto -DH(z)(e_i, 0)\big)$, with D the real Fréchet derivative. For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ the same formula with the complex derivative gives $X_H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$. Thus $X_H = (\partial_I H, -\partial_\theta H)$.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.hamiltonianVectorField`$\mathbb{C}$
Source: HL19 Rem. 2.1, p. 6.

Definition 1.4 (Real Hamiltonian, frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and a real phase point (θ, I), $\mathcal{H}_{h,f}(\theta, I) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$ (`realHamiltonian`). The frequency map is $\nabla h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)\, e_i$ for real I (`frequencyMap`), a real function $\mathbb{R}^n \to \mathbb{R}^n$.
Frozen Lean declaration: `KAM.realHamiltonian, KAM.frequencyMap`
Source: see [K:`bp:p0:nearlyintegrable`].

Definition 1.5 (Torus data on strips)**.** Let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$.

- K is *primary on* T_ρ if $K(\theta + k) = K(\theta) + (k, 0)$ for $\theta \in T_\rho$, $k \in \mathbb{Z}^n$ (`IsPrimaryOnStrip`); *real on reals* if every $K(\theta)$, $\theta \in \mathbb{R}^n$, is a real phase point (`IsRealOnReals`).
- Bilinear dot product $u \cdot v = \sum_i (u_{1,i} v_{1,i} + u_{2,i} v_{2,i})$ on $\mathbb{C}^n \times \mathbb{C}^n$ (`dot`$\mathbb{C}$), $\Omega_0(a, b) = (-b, a)$ (`omegaZero`).
- Invariance error $E_{H,K,\omega}(\theta) = X_H(K(\theta)) - DK(\theta)\omega$ (`invarianceError`).
- Gram matrix $G_K(\theta)_{ij} = DK(\theta)e_i \cdot DK(\theta)e_j$ (`gramMatrix`), a complex $n \times n$ matrix.
- Normal frame $N_K(\theta)_{\cdot j} = \sum_k (G_K^{-1})_{kj}\, \Omega_0\big(DK(\theta)e_k\big) \in \mathbb{C}^n \times \mathbb{C}^n$ (`normalFrame`), with the matrix inverse of Lean.
- Torsion $T_{H,K,\omega}(\theta)_{ij} = N_{\cdot i}(\theta) \cdot \Omega_0\big(DX_H(K(\theta))N_{\cdot j}(\theta) - D_\theta(N_{\cdot j})(\theta)\,\omega\big)$ (`torsion`); $N_{\cdot j}$ is regarded as a function of θ.

Frozen Lean declaration: `KAM.IsPrimaryOnStrip`, `KAM.IsRealOnReals`, `KAM.dotℂ`, `KAM.omegaZero`, `KAM.invarianceError`, `KAM.gramMatrix`, `KAM.normalFrame`, `KAM.torsion`
Source: HL19 (2.2), (3.2)–(3.9), pp. 7, 16–17; see [K:bp:p0:stripdata].

1.3. **Frozen definitions of track I.**

Definition 1.6 (Energy average and extended torsion)**.** For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, K as above, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$, with index set $\{1, \dots, n\} \sqcup \{*\}$:

$$\bar{E}_H(K) = \int_{[0,1]^n} H\big(K(\theta)\big)\, d\theta \in \mathbb{C}, \qquad \mathsf{T}_H(\theta) = \begin{pmatrix} T_{H,K,\omega}(\theta) & \omega \\ \big(DH(K(\theta))N_{\cdot b}(\theta)\big)_b & 0 \end{pmatrix},$$

the $(n+1) \times (n+1)$ complex matrix whose upper-left block is the torsion, last column $(\omega, 0)$ (real entries, viewed in $\mathbb{C}$), and last row $b \mapsto DH(K(\theta))\big(N_{\cdot b}(\theta)\big)$ (complex derivative of H applied to the normal frame), last entry 0. The averaged extended torsion is $\langle \mathsf{T}_H \rangle = \big(\int_{[0,1]^n} \mathsf{T}_H(\theta)_{ij}\, d\theta\big)_{ij}$, the integral taken over real angles.
Frozen Lean declaration: `KAM.Iso.energyAverage`, `KAM.Iso.extendedTorsion`, `KAM.Iso.extendedTorsionAverage`
Source: HL19 (3.13)–(3.14), p. 19, with $c = H$, $d = n$ (Case III), $\hat{\omega} = \omega$; the last row is $DH(K)N$ even off invariance, not $\omega^\top$ (HL19 (3.17) is the invariant case, Rem. 3.7, p. 19).

Definition 1.7 (Bordered Hessian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and real I, the real $(n+1) \times (n+1)$ matrix

$$\mathsf{B}_h(I) = \begin{pmatrix} \big(\partial_b(\nabla h)_a(I)\big)_{a,b} & \nabla h(I) \\ \nabla h(I)^\top & 0 \end{pmatrix},$$

where $\partial_b(\nabla h)_a(I) = D(\nabla h)(I)e_b$ read at a, the real derivative of the frequency map (`borderedHessian`). No symmetry is assumed in the definition.
Frozen Lean declaration: `KAM.Iso.borderedHessian`
Source: Chi05 p. 17 (display after (3.43)); HL19 Rem. 3.7, p. 19.

Definition 1.8 (Hamiltonian solutions with action in B)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $B \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, write $\mathrm{Sol}(H, B, J, z)$ if for all $t \in J$: $z(t)_2 \in B$, H is real differentiable at $z(t)$, and z is differentiable at t with $z'(t) = X_H(z(t))$ (`HasDerivAt`).
Frozen Lean declaration: `KAM.Iso.IsHamiltonianSolutionOn`
Source: ours (D20); the domain B and the differentiability of H are part of the solution concept.

Definition 1.9 (Planar Euclidean length, the primaries, collisions)**.** For $q \in \mathbb{R}^2$: $\rho_2(q)^2 = q_0^2 + q_1^2$ (`radiusSq`) and $\mathrm{rad}(q) = \sqrt{q_0^2 + q_1^2}$ (`radius`; the radicand is nonnegative). For $\mu \in \mathbb{R}$: $a_1(\mu) = (-\mu, 0)$, $a_2(\mu) = (1-\mu, 0)$ (`primary`$_1$, `primary`$_2$). A phase point $z = (q, p) \in \mathbb{R}^2 \times \mathbb{R}^2$ is *collision free* (`CollisionFree` μ) if $\mathrm{rad}(q - a_1(\mu)) > 0$ and $\mathrm{rad}(q - a_2(\mu)) > 0$.
Frozen Lean declaration: `KAM.Iso.radiusSq`, `KAM.Iso.radius`, `KAM.Iso.primary`$_1$, `KAM.Iso.primary`$_2$, `KAM.Iso.CollisionFree`
Source: Chi05 (3.34), (3.37), p. 12–13 (barycentric rotating frame, unit separation, total mass one); ours.

Definition 1.10 (Restricted Hamiltonian and field)**.** On collision-free points $z = (q, p)$, with $r_j = \mathrm{rad}(q - a_j(\mu))$:

$$\mathcal{H}_\mu^{\mathrm{R3}}(q,p) = \tfrac{1}{2}|p|_2^2 + q_1 p_0 - q_0 p_1 - \frac{1-\mu}{r_1} - \frac{\mu}{r_2}, \qquad |p|_2^2 := p_0^2 + p_1^2,$$
$$V_\mu(q,p) = \Big((p_0 + q_1,\ p_1 - q_0),\ (p_1, -p_0) - \frac{1-\mu}{r_1^3}(q - a_1) - \frac{\mu}{r_2^3}(q - a_2)\Big).$$

Thus $V_\mu = (\partial_p \mathcal{H}_\mu^{\mathrm{R3}}, -\partial_q \mathcal{H}_\mu^{\mathrm{R3}})$ on the open collision-free set. A function $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ is a *restricted solution on* J, $\mathrm{RSol}_\mu(J, z)$, if for every $t \in J$ the point $z(t)$ is collision free and z is differentiable at t with $z'(t) = V_\mu(z(t))$.

Frozen Lean declaration: `KAM.Iso.restrictedHamiltonian`, `KAM.Iso.restrictedVectorField`, `KAM.Iso.IsRestrictedSolutionOn`
Source: Chi05 (1.2), p. 2 and (3.34)–(3.39), pp. 12–14, in barycentric rotating coordinates with canonical momentum (not rotating velocity); the equations V_μ are the Hamilton equations of $\mathcal{H}_\mu^{\mathrm{R3}}$ (the check is part of the proofs, see PLAN.md).

Definition 1.11 (Near-circular region)**.** For $a_-, a_+, e_-, e_+ \in \mathbb{R}$, $\mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ is the set of (q, p) for which there exist $r, a, e \in \mathbb{R}$ with

$$r > 0,\ r^2 = q_0^2 + q_1^2, \quad a > 0, \quad 0 < e < 1, \quad a_- < a < a_+, \quad e_- < e < e_+,$$
$$\tfrac{1}{2}|p|_2^2 - \frac{1}{r} = -\frac{1}{2a}, \qquad q_0 p_1 - q_1 p_0 > 0, \qquad (q_0 p_1 - q_1 p_0)^2 = a(1 - e^2).$$

(`NearCircular`.) The three numbers r, a, e are determined by (q, p), so no choice is involved.
Frozen Lean declaration: `KAM.Iso.NearCircular`
Source: Chi05 (2.24)–(2.26), p. 9: $E = -1/(2a)$, $\Gamma^2 = a(1 - e^2)$ in the units $\mu = M = 1$; referred to $\mu = 0$ (primary at the origin).

Definition 1.12 (The Kepler bordered matrix)**.** For $\kappa, \Omega \in \mathbb{R}$ and $L > 0$,

$$\mathsf{K}_{\kappa,\Omega}(L) = \begin{pmatrix} -3\kappa/L^4 & 0 & \kappa/L^3 \\ 0 & 0 & -\Omega \\ \kappa/L^3 & -\Omega & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3},$$

the bordered Hessian $\left(\begin{smallmatrix} D^2 h & \nabla h \\ \nabla h^\top & 0 \end{smallmatrix}\right)$ of $h(L, G) = -\kappa/(2L^2) - \Omega G$ (`keplerBorder`, L in the subtype $\{L : \mathbb{R} \mid L > 0\}$).
Frozen Lean declaration: `KAM.Iso.keplerBorder`
Source: Chi05 p. 17 ($\kappa = \Omega = 1$).

1.4. **Lean totality.**

Remark 1.13 (Lean totality). The frozen definitions use `fderiv` (zero where a function is not differentiable), the matrix inverse (zero for a singular matrix), the Bochner integral (zero for a non-integrable function), division, `Real.sqrt` (zero for a negative argument) and real powers. The statements guard them as follows; each proof must establish the guarded facts before using the defining formulas.

- (T1) *Derivatives of H in I-G1.* D is open, H is holomorphic on D, so DH, X_H, DX_H and D^2H are genuine derivatives at every point of D; K is holomorphic on the open strip T_ρ and $B_\mu(K(\theta)) \subset D$, so $H \circ K$, $X_H \circ K$ and DK are genuine on T_ρ.
- (T2) *Inverses in I-G1.* The Gram matrix inverse is used only under the hypothesis $\det G_K(\theta) \in \mathbb{C}^\times$ on T_ρ, and the inverse of the averaged extended torsion only under $\det\langle \mathsf{T}_H \rangle \in \mathbb{C}^\times$. Then N_K is genuine and holomorphic on T_ρ (by the holomorphic inverse, [K:`bp:p1:holomorphic-inverse`]), so $DN_{\cdot j}$ and T are genuine. For the conclusion, invertibility of the Gram determinant for K_∞ must be proved (it is not a hypothesis on K_∞) before $E_{H,K_\infty,s_\infty\omega^*}$ is used; but note that $E = 0$ and $H \circ K_\infty = E_0$ use only DK_∞ and X_H, not the frame.
- (T3) *Integrals in I-G1.* The cube $[0, 1]^n \subset \mathbb{R}^n \subset T_\rho$ is compact, $\rho > 0$, and the integrands $H \circ K$ and the entries of T_H are continuous on it by (T1)–(T2), hence integrable ([K:`bp:p1:ext-integral`]). Integrability is not a hypothesis; it must be proved.
- (T4) *Powers and division in I-G1.* $\gamma > 0$ and $\rho > 0$, so γ^4, γ^2, $\rho^{4\tau}$, $\rho^{2\tau}$ are positive and the quotients are genuine; $\gamma/(\sum|k_i|)^\tau$ has a positive base since $k \neq 0$.
- (T5) *I-HEAD.* h is holomorphic on the open set B_{r_0}, so ∇h and $D(\nabla h)$ are genuine at real points of B_{r_0} (an open subset of $\mathbb{R}^2$) and B_h is genuine. The solution predicate Sol includes differentiability of $\mathcal{H}_{h,f}$ at each visited point, which also guards $X_{\mathcal{H}}$; it holds for $z(t)_2 \in B \subset B_{r_0}$ and real θ because f is holomorphic on $T_{s_0} \times B_{r_0}$.

(T6) *I-KEPLER, I-R3BP.* $L > 0$ guards every denominator of $\mathsf{K}_{\kappa,\Omega}(L)$. Every inverse distance in $\mathcal{H}^{\mathrm{R3}}_{\mu}$, V_μ is guarded by collision freedom, and the radicand of `radius` is a sum of squares. In $\mathcal{N}$, $r > 0$ and $a > 0$ precede the energy equation.

1.5. **The frozen theorems.**

Theorem 1.14 (I-G1: isoenergetic a-posteriori KAM)**.** *Let $n \geq 2$ and $\tau \in \mathbb{R}$ with $\tau \geq n-1$; let $\rho, \rho_\infty \in \mathbb{R}$ with $0 < \rho_\infty < \rho$; let $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma, \sigma_\omega, \nu \in \mathbb{R}$ with $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma > 0$, $\sigma_\omega > 1$, $\nu > 0$; let $\omega^* \in \mathbb{R}^n$, $\omega^* \neq 0$. Then there exist $C_1, C_2, C_3 > 0$ such that for all $D \subset \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, and $s, \gamma, e, E_0 \in \mathbb{R}$ satisfying*

(I1) D is open and $(\theta + k, I) \in D$ whenever $(\theta, I) \in D$, $k \in \mathbb{Z}^n$;
(I2) H is holomorphic on D, and $H(\theta + k, I) = H(\theta, I)$ on D for $k \in \mathbb{Z}^n$;
(I3) for every real phase point (θ, I) lying in D, $\operatorname{Im} H(\theta, I) = 0$;
(I4) for every $z \in D$: $\|DH(z)\| \leq M$, $\|X_H(z)\| \leq M$, $\|D(X_H)(z)\| \leq M$, $\|D^2(X_H)(z)\| \leq M$, $\|D^2 H(z)\| \leq M$;
(I5) K is holomorphic on T_ρ, primary on T_ρ, and real on reals;
(I6) for every $\theta \in T_\rho$: $B_\mu(K(\theta)) \subset D$ (open max-norm ball) and $\|DK(\theta)\| \leq \sigma_K - \mu$;
(I7) for every $\theta \in T_\rho$: $\det G_K(\theta) \neq 0$ and $|G_K(\theta)^{-1}| \leq \sigma_B - \mu$ (maximal row sum);
(I8) $\det\langle \mathsf{T}_H \rangle \neq 0$ with T_H built from $(H, K, s\omega^*)$, and $|\langle \mathsf{T}_H \rangle^{-1}| \leq \sigma_T - \mu$ (maximal row sum);
(I9) $1 < s < \sigma_\omega$, $\nu \leq (s-1)\|\omega^*\|$, $\nu \leq (\sigma_\omega - s)\|\omega^*\|$;
(I10) $0 < \gamma \leq \bar\gamma$, $\omega^* \in \mathcal{D}_{\gamma,\tau}$, $0 \leq e$;
(I11) $\|E_{H,K,s\omega^*}(\theta)\| \leq e$ for every $\theta \in T_\rho$;
(I12) $|\bar E_H(K) - E_0| \leq e$;
(I13) $C_1 e/(\gamma^4 \rho^{4\tau}) < 1$,

there are $K_\infty : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ and $s_\infty \in \mathbb{R}$ with $1 < s_\infty < \sigma_\omega$ such that

(O1) K_∞ is holomorphic on T_{ρ_∞}, primary on T_{ρ_∞} and real on reals;
(O2) $E_{H,K_\infty,s_\infty\omega^*}(\theta) = 0$ for all $\theta \in T_{\rho_\infty}$;
(O3) $H(K_\infty(\theta)) = E_0$ for all $\theta \in T_{\rho_\infty}$ (pointwise);
(O4) $B_{\mu/2}(K_\infty(\theta)) \subset D$ and $\|DK_\infty(\theta)\| \leq \sigma_K - \mu/2$ for $\theta \in T_{\rho_\infty}$;
(O5) $\|K_\infty(\theta) - K(\theta)\| \leq C_2 e/(\gamma^2 \rho^{2\tau})$ for $\theta \in T_{\rho_\infty}$;
(O6) $\|s_\infty \omega^* - s\omega^*\| \leq C_3 e/(\gamma^2 \rho^{2\tau})$.

Here C_1, C_2, C_3 are chosen after all of $n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma, \sigma_\omega, \nu, \omega^*$ and before $D, H, K, s, \gamma, e, E_0$; hence they may depend on the former, and on none of the latter.

Frozen Lean declaration: `KAM.Iso.isoenergetic_kam`

Source: HL19 Theorem 3.6 and Rem. 3.7, pp. 18–19, in the canonical case $d = n$, $c = h$; proof pp. 45–53. Deviations (all conservative, D-I-G1-UNIFORM, D-I-ENERGY, notes.md): ray data, $\bar\gamma$ and the margins are fixed first; the strict inequalities of (3.16) are replaced by $\leq$ (candidate erratum: with $E_c = 0$ the strict bounds read $0 < 0$); the energy conclusion is pointwise.

Remark 1.15 (Reading of I-G1). (a) The frequency of the approximate torus is $\omega = s\omega^*$ on the open ray $\Theta = \{s'\omega^* : 1 < s' < \sigma_\omega\}$. The Diophantine condition is imposed on ω^*, hence (by homogeneity of $k \mapsto k \cdot \omega$) $s'\omega^*$ is nonresonant for every $s' > 0$, with constant $\gamma s'$. (b) ν enters only through (I9) and through the choice of C_1, C_2, C_3; it measures the distance of s to $\partial\{1 < s' < \sigma_\omega\}$ in the metric of $\mathbb{R}^n$ (HL19 H'_5: $\operatorname{dist}(\omega, \partial\Theta) > 0$ made quantitative, $\operatorname{dist}(s\omega^*, \partial\Theta) \geq \nu$). (c) Taking $e = 0$ gives the exact-torus case; then (I13) holds for every C_1, and (O5), (O6) force $K_\infty = K$ on T_{ρ_∞} and $s_\infty = s$. (d) The row-sum bound in (I7)–(I8) is the max-norm operator norm on $\mathbb{C}^n$, $\mathbb{C}^{n+1}$ respectively.

Theorem 1.16 (I-HEAD: all-time action confinement for two degrees of freedom)**.** *Let $B \subset \mathbb{R}^2$ be open, bounded and nonempty; $s_0, r_0 > 0$; h holomorphic on B_{r_0} with $\operatorname{Im} h(I) = 0$ for every real I with $I \in B_{r_0}$ and*

$$\det \mathsf{B}_h(I) \neq 0 \qquad \text{for every real } I \in B_{r_0}.$$

Let $B' \subset \mathbb{R}^2$ be open with $\overline{B'}$ compact and $\overline{B'} \subset B$, and let $\eta > 0$. Then there is $\varepsilon_0 > 0$ such that for every $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with

- *f holomorphic on $T_{s_0} \times B_{r_0}$, $f(\theta + k, I) = f(\theta, I)$ there for $k \in \mathbb{Z}^2$, and $\operatorname{Im} f(\theta, I) = 0$ for real θ and real $I \in B_{r_0}$;*
- *$0 \le \varepsilon < \varepsilon_0$ and $|f(z)| \le \varepsilon$ on $T_{s_0} \times B_{r_0}$,*

and every $z_0 = (\theta_0, I_0) \in \mathbb{R}^2 \times \mathbb{R}^2$ with $I_0 \in B'$, there is $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with $z(0) = z_0$, $\mathrm{Sol}(\mathcal{H}_{h,f}, B, \mathbb{R}, z)$, $\|z(t)_2 - I_0\| < \eta$ for all $t \in \mathbb{R}$, and: for all $a < 0 < b$ and all $y : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with $y(0) = z_0$ and $\mathrm{Sol}(\mathcal{H}_{h,f}, B, (a, b), y)$ one has $y(t) = z(t)$ for all $t \in (a, b)$.

Frozen Lean declaration: `KAM.Iso.all_time_confinement`

Source: ours (D-I-CONFINEMENT): assembled from I-G1, separation in an energy cylinder (Arnold-type barriers; cf. CC03 Theorem 1.1, p. 7, a numerical one-energy precedent with a truncated potential, not used) and ODE continuation.

Remark 1.17 (Reading of I-HEAD). (a) The constant ε_0 depends on B, s_0, r_0, h, B', η only (it may not depend on f, ε, z_0). (b) $\mathsf{B}_h(I)$ is the matrix of Definition 1.7 with $n = 2$; its determinant is the isoenergetic nondegeneracy of HL19 H'_4 at the torus $\theta \mapsto (\theta, I)$: for $H = h$, $K(\theta) = (\theta, I)$, $\omega = \nabla h(I)$ one computes $N_{\cdot b} = (0, e_b)$, $T = D(\nabla h)(I)$, last row $\nabla h(I)^\top$, so $\mathsf{T}_h = \mathsf{B}_h(I)$ exactly (to be proved in part p1, see PLAN.md). (c) The conclusion contains global existence for all real t and uniqueness among every solution on any interval $(a, b) \ni 0$ that remains in $\{I \in B\}$; in particular it contains uniqueness of z among solutions confined to B. (d) The norm $\|\cdot\|$ is the max norm on $\mathbb{R}^2$.

Theorem 1.18 (I-KEPLER: the classical isoenergetic determinant)**.** *For all $\kappa, \Omega > 0$ and $L > 0$: $\det \mathsf{K}_{\kappa,\Omega}(L) = 3\kappa\Omega^2/L^4$ and $\det \mathsf{K}_{\kappa,\Omega}(L) > 0$.*

Frozen Lean declaration: `KAM.Iso.kepler_bordered_det`

Source: Chi05 p. 17 ($\kappa = \Omega = 1$: $3/L^4$); general κ, Ω is the same cofactor computation (ours).

Theorem 1.19 (I-R3BP: near-circular region of the restricted three-body problem)**.** *Let $a_-, a_+, e_-, e_+ \in \mathbb{R}$ with $0 < a_- < a_+$, $0 < e_- < e_+ < 1/4$ and $a_+(1 + e_+) < 1$. Then the set $\mathcal{N} = \mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ is open and nonempty, and there are $\mu_0, d, R \in \mathbb{R}$ with $0 < \mu_0 \le 1/2$, $d > 0$, $R > 0$ such that for every $\mu \in [0, \mu_0)$ and every $z_0 \in \mathcal{N}$ there is $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with $z(0) = z_0$, $\mathrm{RSol}_\mu(\mathbb{R}, z)$ and for every $t \in \mathbb{R}$*

$$\mathrm{rad}\big(q(t) - a_1(\mu)\big) \ge d, \quad \mathrm{rad}\big(q(t) - a_2(\mu)\big) \ge d,$$
$$\mathrm{rad}(q(t)) \le R, \quad \mathrm{rad}(p(t)) \le R, \qquad z(t) = (q(t), p(t)),$$

and: for all $a < 0 < b$ and y with $y(0) = z_0$ and $\mathrm{RSol}_\mu((a, b), y)$, $y(t) = z(t)$ on (a, b). Here μ_0, d, R depend on $a_\pm, e_\pm$ only, not on μ or z_0.

Frozen Lean declaration: `KAM.Iso.restricted_near_circular`

Source: ours (D-I-PHYSICAL), derived from I-HEAD, I-KEPLER and the Delaunay coordinates of Chi05 §2–3, pp. 8–9, 12–17; not the numerical Victoria theorem of CC03.

Remark 1.20 (Reading of I-R3BP). (a) $\mu = 0$ is included (a massless secondary at $(1, 0)$). (b) The hypothesis $a_+(1+e_+) < 1$ says that the apocentre radius $a(1+e)$ of every orbit of the region is below $1-$margin, so the orbits stay inside the orbit of the secondary; $e_+ < 1/4$ and $e_- > 0$ keep the orbits away from the singularities $e = 0$, $e = 1$ of the Delaunay chart. (c) The region is μ-independent and referenced to $\mu = 0$ (the primary at the origin); the primaries are at $a_1(\mu)$, $a_2(\mu)$ in the dynamics. (d) Collision exclusion for all times is part of the conclusion (distances $\ge d$).

2. Extended torsion and the bordered triangular system

This part collects the linear-algebra and first-order geometry of the isoenergetic scheme of HL19 §6 (pp. 45–51), for the frozen objects $\bar{E}_H$, T_H, $\langle \mathsf{T}_H \rangle$ and B_h of Definitions 1.6 and 1.7. It proves five things.

(1) $\mathsf{T}_H = \mathsf{B}_h(I)$ exactly at the trivial torus $\theta \mapsto (\theta, I)$ (Lemma 2.6); this is why the nondegeneracy hypothesis of I-HEAD is the isoenergetic condition of I-G1.
(2) The last row of T_H is $\omega^\top - (\eta^L)^\top$ exactly, hence $\omega^\top$ on an invariant torus (Lemma 2.8); HL19 Rem. 3.7 only states the invariant case, with a sign slip.
(3) The bordered triangular system of HL19 Lemma 6.1 with its estimates (Lemma 2.12), and its exact origin as the reduction of the linearized isoenergetic equation in the frame (Lemma 2.14). *This is the only genuinely new analytic lemma*; HL19 print only "analogous to Lemma 5.1" for its proof, so the full proof is supplied.
(4) The approximate energy identities (Lemma 2.9): $D(H \circ K) = DH(K)DK$, $\partial_\omega(H \circ K) = -DH(K)E$, the compatibility $\langle \eta^N \rangle = 0$, and the size of the energy defect.
(5) Energy conservation of an invariant torus (Lemma 2.10): $E = 0$ and $\langle H \circ K \rangle = E_0$ give $H \circ K = E_0$ pointwise, the pointwise energy conclusion (O3) of I-G1.

Everything about the frame $L = DK$, N, P, Q, the symplectic defects and the cohomological equation is *cited* from track K's Part 2 ([K:`bp:p2:frame-identities`], [K:`bp:p2:exactness`], [K:`bp:p2:geometry`], [K:`bp:p2:cohomological`], [K:`bp:p2:triangular`], ...) and Part 1, not reproved: the canonical Case III objects of the frozen definitions are those of [K:`bp:p2:objects`] with h replaced by H.

2.1. Setting, sign dictionary, external facts.

Standing assumption 2.1 (Standing setting of Part 2)**.** Let $n \geq 2$ and $\tau \geq 0$ (I-G1 has $\tau \geq n-1$; only $\tau \geq 0$ is used here), $M > 0$. Let $D \subset \mathbb{C}^n \times \mathbb{C}^n$ be open with $(\theta + k, I) \in D$ for $(\theta, I) \in D$, $k \in \mathbb{Z}^n$ (I1); let $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ be holomorphic on D, 1-periodic in θ, real at real phase points of D (I2, I3), with $\|DH\|, \|X_H\|, \|DX_H\|, \|D^2 X_H\| \leq M$ on D (I4). Let $\gamma > 0$ and $\omega \in \mathbb{R}^n \cap \mathcal{D}_{\gamma,\tau}$. Let $r > 0$ and let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ be holomorphic, primary and real on reals on T_r, with $K(T_r) \subset D$ (I5, I6). Where the frame is used, $\det G_K \neq 0$ on T_r.

This is exactly Assumption [K:`bp:p2:standing`] with $h := H$, $\mathfrak{B} := D$, together with the bound $\|DH\| \leq M$, which is used only in Lemma 2.9(e) and Lemma 2.5(b). All objects $L, G, B, N, P, X_N, T, E, A, \Omega_K, E_{\mathrm{sym}}, Q, \eta$ and the averages are those of [K:`bp:p2:objects`], with $h := H$; [K:`bp:p2:frozen-agreement`] identifies them with the frozen `gramMatrix`, `normalFrame`, `torsion`, `invarianceError`. The norms are those of Part 1: max norms, row-sum operator norms $|M| = \max_i \sum_j |M_{ij}|$ (so a $1 \times n$ matrix has the norm $\sum_j |M_{1j}|$, which is the norm of $DH(z)$ in (I4)). Write $\Gamma_\delta := c_R \gamma^{-1} \delta^{-\varkappa}$, $\varkappa = \tau + n$, as in [K:`bp:p2:constants`].

Remark 2.2 (Rays: $s\omega^*$ is Diophantine with the constant of ω^*). In I-G1, $\omega = s\omega^*$ with $s > 1$ and $\omega^* \in \mathcal{D}_{\gamma,\tau}$. For $k \neq 0$, $|k \cdot s\omega^*| = s|k \cdot \omega^*| \geq \gamma / \big(\sum_i |k_i|\big)^\tau$, so $\omega \in \mathcal{D}_{\gamma,\tau}$ with the *same* γ. Hence every result of K's Part 1 and Part 2 that requires $\omega \in \mathcal{D}_{\gamma,\tau}$ applies to $\omega = s\omega^*$ with Γ_δ as above.

Remark 2.3 (Lean totality for this part). Following Remark 1.13: every statement below that uses N, T, T_H assumes $\det G_K \neq 0$ on T_r first, so that $B = G^{-1}$ is the true inverse, holomorphic ([K:`bp:p1:holomorphic-inverse`]); every use of $\langle \mathsf{T}_H \rangle$ is preceded by continuity of T_H on $\mathbb{R}^n$ (Lemma 2.5(a)), hence integrability ([K:`bp:p1:ext-integral`]); every derivative of H, X_H is taken at a point of the open set D where H is holomorphic, so `fderiv` is the genuine derivative; the frozen T_H entry `fderiv` $\mathbb{C}$ H $(K\theta)$ $(N_{\cdot b})$ is then the matrix product $DH(K(\theta))N_{\cdot b}(\theta)$.

Remark 2.4 (Sign dictionary HL19 $\leftrightarrow$ this blueprint). HL19 use $\mathfrak{L}_\omega = -\sum_i \omega_i \partial_{\theta_i} = -\partial_\omega$ (4.1), p. 20, and the right operator $\mathfrak{R}_\omega$ of (4.2), $\widehat{(\mathfrak{R}_\omega v)}_k = -\hat{v}_k/(2\pi i k \cdot \omega)$. Track K's operator $\mathcal{R}$ ([K:`bp:p2:cohomological`]) solves $\partial_\omega u = g$, $\langle u \rangle = 0$, i.e. $\widehat{(\mathcal{R}g)}_k = \hat{g}_k/(2\pi i k \cdot \omega)$. Thus

$$\mathfrak{L}_\omega = -\partial_\omega, \qquad \mathfrak{R}_\omega = -\mathcal{R} \quad \text{(on the data of (4.1))}.$$

With the canonical $\Omega = \Omega_0$ and $\Omega X_h = (Dh)^\top$ (Rem. 2.1, p. 6; our [K:`bp:p2:hamiltonian`](d)), HL19's invariance error is $E = X_h(K) - DK\,\omega$ (2.2), p. 7, which p. 45 writes as $X_h(K) + \mathfrak{L}_\omega K$.

The linearized system (6.1), p. 45, with $\bar\omega = \omega + \Delta\omega$, $\Delta\omega = -\omega\xi^\omega$ (6.3), is therefore

$$A\,\Delta K - \partial_\omega \Delta K - DK\,\Delta\omega = A\,\Delta K - \partial_\omega \Delta K + L\omega\,\xi^\omega = -E, \qquad \langle DH(K)\Delta K\rangle = -E^\omega,$$

$A = DX_H \circ K$, $L = DK$, $E^\omega = \langle H\circ K\rangle - E_0$, and $\bar\omega = (1-\xi^\omega)\omega$ as on p. 50. The system (6.7) is

$$T\xi^N - \partial_\omega \xi^L + \omega\,\xi^\omega = \eta^L, \qquad -\partial_\omega \xi^N = \eta^N, \qquad \langle Y\xi^N\rangle = \eta^\omega,$$

and the solution (6.8)–(6.10) reads $\xi^N = \xi_0^N - \mathcal{R}\eta^N$, $\xi^L = \xi_0^L + \mathcal{R}\big(T\xi^N + \omega\xi^\omega - \eta^L\big)$, with (6.10) unchanged because $\mathfrak{R}_\omega = -\mathcal{R}$. Lemma 2.14 verifies that this is the equation obtained from (6.1) in the frame, signs included. Here and below $Y(\theta) := DH(K(\theta))N(\theta) \in \mathbb{C}^{1\times n}$ is HL19's H of (6.14), renamed to avoid a clash with the Hamiltonian.

External facts used in this part. Besides [K:`bp:p1:ext-integral`] (continuous functions on a compact box are integrable, integrals are linear and bounded by the supremum), the proofs use only:

(X1) *Chain and product rules* for Fréchet derivatives of holomorphic maps, and linearity of the derivative (Mathlib `HasFDerivAt.comp`, `fderiv_add`, `fderiv_const`).

(X2) *Complex derivative in a real direction.* If φ is holomorphic on an open $U \subset \mathbb{C}^n$, $x \in U \cap \mathbb{R}^n$ and $v \in \mathbb{R}^n$, then $\varphi|_{U\cap\mathbb{R}^n}$ has the real directional derivative $D\varphi(x)v$ at x; if moreover φ is real on $U \cap \mathbb{R}^n$, then $D\varphi(x)v \in \mathbb{R}$ (it is a limit of real difference quotients). See [K:`bp:p1:hamiltonian-real`], step [K:`step:p1:hr-derivative`].

2.2. **The extended torsion.**

Lemma 2.5 (Regularity, periodicity, realness and size of the extended torsion)**.** *Assume 2.1 with* $\det G_K \neq 0$ *on* T_r*, and put* $Y := DH(K)N$ *on* T_r*. Then:*

(a) Y *and the entries of* T_H *(built from* (H,K,ω)*) are holomorphic and* 1*-periodic on* T_r *and real at real* θ*. In particular* T_H *is continuous on* $\mathbb{R}^n$ *and* $\langle \mathsf{T}_H\rangle$ *is defined, real, and equals the block matrix*

$$\langle \mathsf{T}_H\rangle = \begin{pmatrix} \langle T\rangle & \omega \\ \langle Y\rangle & 0 \end{pmatrix}$$

with $\langle T\rangle = \langle T_{H,K,\omega}\rangle$ *the torsion average of* [K:`bp:p2:objects`].

(b) If $|DK| \le \sigma_K'$ *and* $|B| \le \sigma_B'$ *on* T_r*, then* $|Y| \le M\sigma_K'\sigma_B'$ *on* T_r*, and on any set where* $|T| \le C_T$*,* $|\mathsf{T}_H| \le \max\{C_T + |\omega|,\ M\sigma_K'\sigma_B'\}$ *in the row-sum norm of* $\mathbb{C}^{(n+1)\times(n+1)}$*.*

Frozen Lean declaration: `KAM.Iso.extendedTorsion, KAM.Iso.extendedTorsionAverage`
Source: HL19 (3.13), p. 19 ($c = H$*,* $d = n$*); HL19 (6.14), p. 47 for* Y*. Estimates: ours.*

Proof. **Step S2.1: Holomorphy.** N is holomorphic on T_r ([K:`bp:p2:frame-basic`](a)), $K(T_r) \subset D$ and DH is holomorphic on D ([K:`bp:p1:holomorphic-regularity`]), so $Y = DH \circ K \cdot N$ is a product of holomorphic functions ((X1)). The entries of T and the constant block ω are holomorphic by [K:`bp:p2:frame-basic`](a). Hence every entry of T_H is holomorphic, in particular continuous on $\mathbb{R}^n \subset T_r$.

Step S2.2: Periodicity and realness. Differentiating $H(\theta+k, I) = H(\theta, I)$ gives $DH(z+(k,0)) = DH(z)$ on D, and $K(\theta+k) = K(\theta) + (k,0)$; so $DH(K(\theta+k)) = DH(K(\theta))$. N and T are 1-periodic by [K:`bp:p2:periodicity`]. At real θ, $K(\theta)$ is a real phase point of D, where DH is real (the argument of [K:`bp:p2:hamiltonian`](c), step [K:`step:p2:ham-real`], applies to H, real at real points), and N, T are real ([K:`bp:p2:periodicity`]). Hence Y is real at real points.

Step S2.3: The average. By Step S2.1, each entry is integrable on $[0,1]^n$ ([K:`bp:p1:ext-integral`]), so the Bochner integral of the frozen $\langle \mathsf{T}_H\rangle$ is the entrywise integral, real because the integrand is real at real θ; the entries of the border are $\langle\omega\rangle = \omega$ (the cube has volume one) and $\langle 0\rangle = 0$. (Lean totality, Remark 2.3.)

Step S2.4: Bounds. $|Y| \le |DH(K)||N| \le M\,|\Omega_0||L||B| \le M\sigma'_K\sigma'_B$, by [K:bp:p2:matrix](b), (d). The last column of T_H is the $n\times 1$ matrix ω, of norm $|\omega|$; by [K:bp:p2:matrix](c) the $(n+1)\times(n+1)$ norm is at most the maximum of the two block-row bounds $C_T + |\omega|$ and $|Y|$. □

Lemma 2.6 ($\mathsf{T}_H = \mathsf{B}_h$ at the trivial torus)**.** *Let $U \subset \mathbb{C}^n$ be open and $h : U \to \mathbb{C}$ holomorphic with $h(J) \in \mathbb{R}$ for every real $J \in U$. Let $D = \mathbb{C}^n \times U$, $H(\theta, J) := h(J)$ on D, $I \in U \cap \mathbb{R}^n$ and $K(\theta) = (\theta, I)$ ($\theta \in \mathbb{C}^n$). Put $\omega := \nabla h(I) \in \mathbb{R}^n$ (`frequencyMap`). Then K is holomorphic, primary and real on reals, $K(\mathbb{C}^n) \subset D$, $G_K = I_n$, $E_{H,K,\omega} \equiv 0$, and for every $\theta \in \mathbb{C}^n$*

$$\mathsf{T}_H(\theta) = \mathsf{B}_h(I) \quad \textit{(the real matrix, viewed in } \mathbb{C}^{(n+1)\times(n+1)}\textit{)}, \qquad \langle \mathsf{T}_H \rangle = \mathsf{B}_h(I),$$

the frozen `borderedHessian`. In particular $\det\langle \mathsf{T}_H\rangle \neq 0$ iff $\det \mathsf{B}_h(I) \neq 0$ and $|\langle \mathsf{T}_H\rangle^{-1}| = |\mathsf{B}_h(I)^{-1}|$.
Frozen Lean declaration: `KAM.Iso.extendedTorsion`, `KAM.Iso.extendedTorsionAverage`, `KAM.Iso.borderedHessian`
Source: ours; HL19 Rem. 3.7, p. 19, and Chi05 p. 17 (determinant $3/L^4$ in the Kepler case). Reading remark: Remark 1.17(b).

Proof. **Step S2.5: The torus.** $K(\theta+k) = (\theta+k, I) = K(\theta)+(k,0)$, K is affine, hence holomorphic, and real on real θ because I is real. $DK(\theta)e_i = (e_i, 0)$, so $G_K = (e_i \cdot e_j) = I_n$ (using the bilinear dot of Definition 1.5), $\det G_K = 1$ and $B = I_n$.

Step S2.6: The normal frame. $N_{\cdot j} = \sum_k \delta_{kj}\Omega_0(e_k, 0) = \Omega_0(e_j, 0) = (0, e_j)$, a constant function of θ, so $D_\theta N_{\cdot j} = 0$.

Step S2.7: The vector field. Since $H(\theta, J) = h(J)$, $X_H(\theta, J) = \big((\partial_a h(J))_a,\, 0\big)$, so $E_{H,K,\omega}(\theta) = X_H(K\theta) - DK\,\omega = (\nabla^{\mathbb{C}} h(I) - \omega, 0)$, where $\nabla^{\mathbb{C}} h(I) = (\partial_a h(I))_a$. By (X2) applied to $\varphi = h$ (real on $U \cap \mathbb{R}^n$), $\partial_a h(J) \in \mathbb{R}$ for real J, and the real function $J \mapsto \operatorname{Re} h(J) = h(J)$ on $U \cap \mathbb{R}^n$ has partial derivative $\partial_a h(J)$ in the direction e_a; by the definition of `frequencyMap`, $\omega_a = \partial_a h(I)$. Hence $E \equiv 0$, and more generally $\nabla h(J)_a = \partial_a h(J)$ for real $J \in U$.

Step S2.8: The torsion. $DX_H(\theta, I)(0, e_j) = \big((\partial_j \partial_a h(I))_a, 0\big)$, and $\Omega_0(w, 0) = (0, w)$. Hence $T(\theta)_{ij} = N_{\cdot i} \cdot \Omega_0\big(DX_H N_{\cdot j} - 0\big) = (0, e_i) \cdot (0, (\partial_j \partial_a h(I))_a) = \partial_j \partial_i h(I)$.

Step S2.9: The border. The last column is $\omega = \nabla h(I)$, entries $\omega_a = \partial_a h(I)$. The last row is $DH(K\theta)N_{\cdot b} = DH(\theta, I)(0, e_b) = \partial_b h(I) = \omega_b$. The corner is 0.

Step S2.10: Comparison with the bordered Hessian. The (a, b) entry of $\mathsf{B}_h(I)$ is $D(\nabla h)(I)e_b$ read at a, the real derivative in direction e_b of $J \mapsto \nabla h(J)_a = \partial_a h(J)$ on $U \cap \mathbb{R}^n$ (Step S2.7); since $\partial_a h$ is holomorphic, this is $\partial_b \partial_a h(I)$ by (X2), a real number. So the upper left blocks agree (and are symmetric by [K:bp:p1:holomorphic-regularity]); the border of $\mathsf{B}_h(I)$ is $\nabla h(I)$ in the last column and row, as found above. Thus $\mathsf{T}_H(\theta) = \mathsf{B}_h(I)$ for all θ; the constant has average equal to itself since $[0, 1]^n$ has volume one. The statements on the determinant and the inverse follow because a real matrix and its image in $\mathbb{C}^{(n+1)\times(n+1)}$ have the same determinant and inverse ([K:bp:p1:real-inverse]) and the same row-sum norm. □

Remark 2.7 (Use). For $H = h + f$ with f small, the torus $K(\theta) = (\theta, I)$ with $\omega = \nabla h(I)$ (or $s\omega^*$ with a nearby I, see Part 3) has $\langle \mathsf{T}_H \rangle = \mathsf{B}_h(I) + O(\|f\|)$ by Lemma 2.8 below and the Cauchy estimates; this is how (I8) of I-G1 is verified from $\det \mathsf{B}_h \neq 0$, with σ_T from [K:bp:p1:neumann].

Lemma 2.8 (The energy row of the extended torsion)**.** *Assume 2.1 with $\det G_K \neq 0$ on T_r and let $E = E_{H,K,\omega}$, $\eta^L = -N^\top \Omega_0 E$ as in* [K:bp:p2:objects]. *Then on T_r:*

(a) $Y = DH(K)N = \omega^\top - (\eta^L)^\top$. In particular $|Y - \omega^\top| \le |\eta^L|$ and, entrywise, $Y_b = \omega_b + N_{\cdot b}^\top \Omega_0 E$.

(b) If $E = 0$ on T_r (invariant torus), then $Y = \omega^\top$ on T_r, so

$$\mathsf{T}_H(\theta) = \begin{pmatrix} T(\theta) & \omega \\ \omega^\top & 0 \end{pmatrix} \quad (\theta \in T_r), \qquad \langle \mathsf{T}_H \rangle = \begin{pmatrix} \langle T \rangle & \omega \\ \omega^\top & 0 \end{pmatrix}.$$

(c) *In general* $\langle \mathsf{T}_H \rangle = \mathsf{T}^{\mathrm{inv}} - \begin{pmatrix} 0 & 0 \\ \langle \eta^L \rangle^\top & 0 \end{pmatrix}$, *where* $\mathsf{T}^{\mathrm{inv}} := \left(\begin{smallmatrix} \langle T \rangle & \omega \\ \omega^\top & 0 \end{smallmatrix}\right)$ *is built from the same* T*; and if* $|E| \le e$ *on* T_r, $|L| \le \sigma'_K$, $|B| \le \sigma'_B$, *then* $|\langle \mathsf{T}_H \rangle - \mathsf{T}^{\mathrm{inv}}| \le 2n^2 \sigma'_K \sigma'_B \, e$.

Frozen Lean declaration: `KAM.Iso.extendedTorsion`

Source: HL19 Rem. 3.7, p. 19 (part (b), (3.17)), proved with the sign corrected (Remark 2.15, E3); (a), (c) are ours: HL19 (3.13) defines the last row as $Dc(K)N$ *without simplifying it off invariance.*

Proof. **Step S2.11: The gradient identity.** By [K:`bp:p2:hamiltonian`](d) (equivalently [K:`bp:p1:hamiltonian-real`]), $\Omega_0 X_H = (DH)^\top$ on D, so $DH(z) = (\Omega_0 X_H(z))^\top = X_H(z)^\top \Omega_0^\top = -X_H(z)^\top \Omega_0$.

Step S2.12: Insert the error. At $z = K(\theta)$, $X_H(K) = L\omega + E$ by the definition of E. Hence

$$Y = DH(K)N = -(L\omega + E)^\top \Omega_0 N = -\omega^\top L^\top \Omega_0 N - E^\top \Omega_0 N = \omega^\top - E^\top \Omega_0 N,$$

using $L^\top \Omega_0 N = -I_n$ ([K:`bp:p2:frame-identities`](a)). The b-th entry of the row $E^\top \Omega_0 N$ is the scalar $E^\top \Omega_0 N_{\cdot b} = N_{\cdot b}^\top \Omega_0^\top E = -N_{\cdot b}^\top \Omega_0 E = (\eta^L)_b$. This is (a) and the entrywise form. The row-sum norm of the $1 \times n$ matrix $(\eta^L)^\top$ is $\sum_b |\eta^L_b| \le n|\eta^L|$, so $|Y - \omega^\top| \le n|\eta^L|$.

Step S2.13: Invariant torus and average. (b) is (a) with $\eta^L = 0$ (since $E = 0$), together with the definition of T_H (Definition 1.6) and Lemma 2.5(a). For (c): averaging (a) and using linearity, $\langle Y \rangle = \omega^\top - \langle \eta^L \rangle^\top$, so the two matrices differ only in the last row, which has norm $\le n|\langle \eta^L \rangle| \le n \sup_{\mathbb{R}^n} |\eta^L|$ ([K:`bp:p2:averages`](a)), and $|\eta^L| \le |N^\top||\Omega_0||E| \le 2n\,|N||E| \le 2n\,\sigma'_K \sigma'_B \, e$ ([K:`bp:p2:frame-basic`](c), with $|N| \le |L||B|$). Hence the bound $2n^2 \sigma'_K \sigma'_B e$. □

2.3. **Energy identities.**

Lemma 2.9 (Approximate energy identities)**.** *Assume 2.1 and let* $E = E_{H,K,\omega}$, $E^{\mathrm{en}} := H \circ K - \langle H \circ K \rangle$ *on* T_r*. Then:*

(a) $H \circ K$ *is holomorphic, 1-periodic on* T_r*, real on reals, and* $D(H \circ K) = DH(K)\, DK$*; the average* $\langle H \circ K \rangle = \bar{E}_H(K)$ *is the frozen* `energyAverage`*, real.*

(b) $\partial_\omega (H \circ K) = D(H \circ K)\omega = -DH(K)\, E$ *on* T_r*.*

(c) $L^\top \Omega_0 E = (D(H \circ K))^\top - \Omega_K \omega$ *on* T_r*. If* $\det G_K \neq 0$*, then* $\eta^N = L^\top \Omega_0 E$*.*

(d) $\langle D(H \circ K) \rangle = 0$, $\langle \Omega_K \rangle = 0$ *and* $\langle \eta^N \rangle = 0$ *(the compatibility condition of Lemma 2.12).*

(e) *If* $|E| \le \varepsilon$ *on* T_r *and* $0 < \delta < r$*, then* $|E^{\mathrm{en}}| \le M \Gamma_\delta \varepsilon$ *on* $T_{r-\delta}$*, and for* $0 < \delta' < r - \delta$*,* $|D(H \circ K)| \le nM\Gamma_\delta \varepsilon / \delta'$ *on* $T_{r-\delta-\delta'}$*.*

(f) *(Energy defect.) Let* $\det G_K \neq 0$ *on* T_r*, let* $\xi = (\xi^L, \xi^N)$ *be holomorphic on a strip* $T_{r'}$*,* $r' \le r$*, and* $\Delta K = P\xi$*. Then on* $T_{r'}$

$$DH(K)\, \Delta K = Y \xi^N + D(H \circ K)\, \xi^L = Y \xi^N + (\eta^N)^\top \xi^L - \omega^\top \Omega_K \, \xi^L.$$

Consequently, if $\langle Y \xi^N \rangle = \eta^\omega$*, then* $\langle DH(K) \Delta K \rangle = \eta^\omega + E^\omega_{\mathrm{lin}}$ *with* $E^\omega_{\mathrm{lin}} := \langle D(H \circ K)\, \xi^L \rangle$*, and* $|E^\omega_{\mathrm{lin}}| \le \sup_{\mathbb{R}^n} |D(H \circ K)| \, \sup_{\mathbb{R}^n} |\xi^L| \le n \big(\sup |\eta^N| + |\omega| \sup |\Omega_K| \big) \sup |\xi^L|$.

Frozen Lean declaration: `KAM.Iso.energyAverage`

Source: HL19 Lemma 4.2, (4.3)–(4.5), p. 21, and (4.9) ($\mathfrak{L}_\omega (c \circ K) = Dc(K)E$*, equivalent to (b) with* $\mathfrak{L}_\omega = -\partial_\omega$*); p. 46 (*$Dc(K)L = D(c \circ K)$*,* E^ω_{lin} *of p. 49); (6.11)–(6.14), p. 47; compatibility* $\langle \eta^N \rangle = 0$ *from Lemma 4.6 as recalled on p. 47.*

Proof. **Step S2.14: Regularity and chain rule.** K is holomorphic on T_r with values in D, and H is holomorphic on D, so $H \circ K$ is holomorphic with $D(H \circ K) = DH(K) DK$ ((X1)). It is 1-periodic since $H(K(\theta + k)) = H(K(\theta) + (k, 0)) = H(K(\theta))$, and real at real θ because $K(\theta)$ is a real phase point of D (I3). The average is $\int_{[0,1]^n} H(K(\theta)) d\theta = \bar{E}_H(K)$ by definition, integrable by [K:`bp:p2:averages`](a). This is (a).

Step S2.15: The derivative along ω. $D(H\circ K)\omega = DH(K)L\omega = DH(K)\big(X_H(K) - E\big)$. By [K:bp:p2:hamiltonian](d), $DH = (\Omega_0 X_H)^\top$, so $DH(z)X_H(z) = X_H^\top \Omega_0^\top X_H = -X_H^\top \Omega_0 X_H = 0$ (Ω_0 is antisymmetric). Hence $D(H\circ K)\omega = -DH(K)E$, which is (b).

Step S2.16: The gradient identity. This repeats the gradient step of [K:bp:p2:exactness]: $L^\top \Omega_0 X_H(K) = (DH(K)L)^\top = (D(H\circ K))^\top$, hence $L^\top\Omega_0 E = (D(H\circ K))^\top - L^\top\Omega_0 L\omega = (D(H\circ K))^\top - \Omega_K\omega$. The identification $\eta^N = L^\top\Omega_0 E$ is the definition of η in [K:bp:p2:objects]. This is (c).

Step S2.17: Averages. $D(H\circ K)$ has the entries $\partial_{\theta_\ell}(H\circ K)$ of a holomorphic periodic function, so the average vanishes ([K:bp:p2:averages](b)); $\langle\Omega_K\rangle = 0$ and $\langle\eta^N\rangle = 0$ are [K:bp:p2:exactness]. (Also directly: $\langle\eta^N\rangle = \langle D(H\circ K)\rangle^\top + \langle\Omega_K\rangle\omega = 0$ by (c).)

Step S2.18: The size of the energy fluctuation. E^{en} is holomorphic and 1-periodic on T_r, has zero average, and by (b) $|\partial_\omega E^{\mathrm{en}}| = |DH(K)E| \le |DH(K)||E| \le M\varepsilon$ on T_r (here (I4), $K(T_r)\subset D$, is used). [K:bp:p2:cohomological](b), on T_r with loss δ, gives $|E^{\mathrm{en}}| \le \Gamma_\delta M\varepsilon$ on $T_{r-\delta}$. The Cauchy estimate [K:bp:p1:cauchy-strip] for the bounded holomorphic function E^{en} on $T_{r-\delta}$ with loss δ' gives $|DE^{\mathrm{en}}| = |D(H\circ K)| \le n\Gamma_\delta M\varepsilon/\delta'$ on $T_{r-\delta-\delta'}$. This is (e).

Step S2.19: The energy defect. $P = (L\ N)$, so $\Delta K = L\xi^L + N\xi^N$ and $DH(K)\Delta K = DH(K)L\xi^L + DH(K)N\xi^N = D(H\circ K)\xi^L + Y\xi^N$ by (a). By (c), $D(H\circ K) = (\eta^N + \Omega_K\omega)^\top = (\eta^N)^\top + \omega^\top\Omega_K^\top = (\eta^N)^\top - \omega^\top\Omega_K$ since $\Omega_K^\top = -\Omega_K$ ($\Omega_K = L^\top\Omega_0 L$, $\Omega_0^\top = -\Omega_0$). The average of the sum splits linearly. For the bound: $|\langle D(H\circ K)\xi^L\rangle| \le \sup_{\mathbb{R}^n}|D(H\circ K)\xi^L|$ ([K:bp:p2:averages](a)), and $|D(H\circ K)| \le |(\eta^N)^\top| + |\omega^\top\Omega_K| \le n|\eta^N| + n|\omega||\Omega_K|$, with [K:bp:p2:matrix](a) for transposes (a $1\times n$ row $v^\top$ has $|v^\top| = \sum|v_i| \le n|v|$). □

Lemma 2.10 (Energy conservation of an invariant torus)**.** *Assume 2.1 on a strip T_r (with $\omega\in\mathcal{D}_{\gamma,\tau}$) and $E_{H,K,\omega}(\theta) = 0$ for all $\theta\in T_r$. Then:*

(a) $H\circ K \equiv \bar E_H(K)$ is constant on T_r.
(b) If $E_0\in\mathbb{R}$ (or $\mathbb{C}$) and $\bar E_H(K) = E_0$, then $H(K(\theta)) = E_0$ for every $\theta\in T_r$.
(c) Quantitatively, if $E_{H,K,\omega}(\theta) = 0$ on T_r then $\sup_{T_r}|H\circ K - E_0| = |\bar E_H(K) - E_0|$.

Frozen Lean declaration: `KAM.Iso.energyAverage`
Source: HL19 Rem. 2.2, p. 7 ("quasi-periodicity implies $h(K(\theta)) = \langle h\circ K\rangle$"); here proved. This is the pointwise energy conclusion (O3) of I-G1 (decision D-I-ENERGY).

Proof. **Step S2.20: The derivative vanishes along ω.** Let $u := H\circ K - \langle H\circ K\rangle$, holomorphic and 1-periodic on T_r with $\langle u\rangle = 0$ (Lemma 2.9(a)). By Lemma 2.9(b), $Du\,\omega = -DH(K)E = 0$ on T_r.

Step S2.21: Uniqueness for the cohomological equation. For every $0<\delta<r$, $u|_{T_{r-\delta}}$ is holomorphic, 1-periodic, of zero average, and solves $Du\,\omega = g$ with $g = 0$ (holomorphic, bounded, periodic, $\langle g\rangle = 0$ on T_r). By the uniqueness clause of [K:bp:p1:cohomological] (no boundedness of the solution assumed), $u = 0$ on $T_{r-\delta}$. As $\delta\in(0,r)$ is arbitrary, $u = 0$ on $T_r = \bigcup_\delta T_{r-\delta}$, which is (a).

Step S2.22: Pointwise energy. (b) and (c) are (a): $H\circ K \equiv \bar E_H(K)$, so $|H(K(\theta)) - E_0| = |\bar E_H(K) - E_0|$ for every θ. □

Remark 2.11 (Where (O3) of I-G1 comes from). In the proof of I-G1 (Part 2), $E_{H,K_\infty,s_\infty\omega^*} = 0$ on T_{ρ_∞} and $\bar E_H(K_\infty) = E_0$: the second follows from $|\bar E_H(K_j) - E_0| \to 0$ along the iteration and the uniform convergence $K_j\to K_\infty$ (continuity of H and of the integral on the compact cube), which is part of Part 2; then Lemma 2.10(b), with $\omega = s_\infty\omega^*\in\mathcal{D}_{\gamma,\tau}$ (Remark 2.2), gives $H(K_\infty) = E_0$ on T_{ρ_∞}. Neither the Gram determinant of K_∞ nor the frame is used (Remark 1.13(T2)).

2.4. **The bordered triangular system.** Notation: for $T : T_r \to \mathbb{C}^{n\times n}$ and $Y : T_r \to \mathbb{C}^{1\times n}$ holomorphic and 1-periodic put $\mathcal{T} := \begin{pmatrix} \langle T\rangle & \omega \\ \langle Y\rangle & 0 \end{pmatrix} \in \mathbb{C}^{(n+1)\times(n+1)}$; with $T = T_{H,K,\omega}$, $Y = DH(K)N$, this is $\langle \mathsf{T}_H\rangle$ (Lemma 2.5(a)). Let $\mathcal{R}$ be the solution operator of [K:bp:p2:cohomological](a): for g holomorphic, periodic with $\langle g\rangle = 0$, $\mathcal{R}g$ is the unique zero-average periodic solution of $\partial_\omega u = g$.

Lemma 2.12 (Bordered triangular equations (HL19 Lemma 6.1))**.** *Let $\gamma > 0$, $\tau \ge 0$, $\omega \in \mathbb{R}^n \cap \mathcal{D}_{\gamma,\tau}$, $r > 0$, $\delta > 0$ with $2\delta < r$, and $\Gamma_\delta = c_R\gamma^{-1}\delta^{-\varkappa}$. Let $\eta^L, \eta^N : T_r \to \mathbb{C}^n$ be holomorphic, 1-periodic, $|\eta^L| \le H_L$ and $|\eta^N| \le H_N$ on T_r, with $\langle \eta^N\rangle = 0$, and let $\eta^\omega \in \mathbb{C}$. Let $T : T_r \to \mathbb{C}^{n\times n}$ and $Y : T_r \to \mathbb{C}^{1\times n}$ be holomorphic and 1-periodic with $|T| \le H_T$, $|Y| \le H_Y$ on $T_{r-\delta}$. Assume $\det \mathcal{T} \ne 0$ and $|\mathcal{T}^{-1}| \le \sigma$ (row-sum norm of $\mathbb{C}^{(n+1)\times(n+1)}$). Fix $\xi_0^L \in \mathbb{C}^n$ and define, in this order,*

$$
\begin{aligned}
u^N &:= \mathcal{R}\eta^N \quad \text{on } T_{r-\delta},\\
\begin{pmatrix} \xi_0 \\ \xi^\omega \end{pmatrix} &:= \mathcal{T}^{-1}\begin{pmatrix} \langle \eta^L\rangle + \langle Tu^N\rangle \\ \eta^\omega + \langle Yu^N\rangle \end{pmatrix} \in \mathbb{C}^n \times \mathbb{C},\\
\xi^N &:= \xi_0 - u^N, \qquad g^L := T\xi^N + \omega\,\xi^\omega - \eta^L \quad \text{on } T_{r-\delta},\\
\xi^L &:= \xi_0^L + \mathcal{R}g^L \quad \text{on } T_{r-2\delta}.
\end{aligned}
$$

Then:

(a) *(Well defined.) u^N, ξ^N, g^L are holomorphic and 1-periodic on $T_{r-\delta}$, $\langle g^L\rangle = 0$, so ξ^L is defined and holomorphic, 1-periodic on $T_{r-2\delta}$; the averages $\langle Tu^N\rangle$, $\langle Yu^N\rangle$ exist.*

(b) *(The system.) On $T_{r-2\delta}$,*

$$T\xi^N - \partial_\omega\xi^L + \omega\,\xi^\omega = \eta^L, \qquad -\partial_\omega\xi^N = \eta^N, \qquad \langle Y\xi^N\rangle = \eta^\omega,$$

i.e. $\Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega = \eta$ with $\Lambda = \left(\begin{smallmatrix} 0 & T \\ 0 & 0 \end{smallmatrix}\right)$, $\xi = (\xi^L, \xi^N)$, $\eta = (\eta^L, \eta^N)$.

(c) *(Estimates.) With $S_* := \sigma\max\{H_L + H_T\Gamma_\delta H_N,\ |\eta^\omega| + H_Y\Gamma_\delta H_N\}$:*

$$|\xi_0|,\ |\xi^\omega| \le |(\xi_0, \xi^\omega)| \le S_*, \qquad \sup_{T_{r-\delta}}|\xi^N| \le S_* + \Gamma_\delta H_N,$$

$$\sup_{T_{r-\delta}}|g^L| \le H_T(S_* + \Gamma_\delta H_N) + |\omega|S_* + H_L, \qquad \sup_{T_{r-2\delta}}|\xi^L - \xi_0^L| \le \Gamma_\delta \sup_{T_{r-\delta}}|g^L|.$$

(d) *(Uniqueness.) If $(\tilde\xi^L, \tilde\xi^N, \tilde\xi^\omega)$ are holomorphic, periodic on $T_{r-2\delta}$ ($\tilde\xi^\omega \in \mathbb{C}$) and satisfy the three equations of (b) on $T_{r-2\delta}$, then $\tilde\xi^\omega = \xi^\omega$, $\tilde\xi^N = \xi^N$ and $\tilde\xi^L - \xi^L$ is constant. Thus the solution is unique up to $\langle \xi^L\rangle$ (the free phase direction, HL19 Rem. 3.5).*

(e) *(Realness.) If ω is real and η^L, η^N, T, Y are real on reals, $\eta^\omega \in \mathbb{R}$, $\mathcal{T}$ real (automatic), and $\xi_0^L \in \mathbb{R}^n$, then ξ_0, ξ^ω, ξ^N, ξ^L, g^L are real on reals.*

Source: HL19 Lemma 6.1, pp. 46–47, (6.7)–(6.10), with signs converted by Remark 2.4; the estimates are those on p. 46, with the additional term $|\omega|S_*$ in g^L (see Remark 2.13). Structure as in [K:bp:p2:triangular] *(HL19 Lemma 5.1, pp. 31–32).*

Source gap: HL19 prove Lemma 6.1 in four lines ("It is analogous to Lemma 5.1…"), and Lemma 5.1 is itself only referred to ("directly adapted from Lemma 4.14 in [33]"). The proof below is supplied here in full; the new points relative to [K:bp:p2:triangular] *are Steps S2.25, S2.26 and S2.27, and uniqueness (d).*

Proof. **Step S2.23: First cohomological equation.** η^N is holomorphic, periodic on T_r, bounded by H_N, with $\langle \eta^N\rangle = 0$. [K:bp:p2:cohomological](a) gives $u^N = \mathcal{R}\eta^N$ holomorphic and 1-periodic on $T_{r-\delta}$, $\langle u^N\rangle = 0$, $\partial_\omega u^N = \eta^N$, $|u^N| \le \Gamma_\delta H_N$ on $T_{r-\delta}$, real on reals if η^N is.

Step S2.24: Averages of the products. Tu^N and Yu^N are holomorphic and periodic on $T_{r-\delta}$, hence continuous on $\mathbb{R}^n$, so their averages exist ([K:bp:p2:averages](a)) and $|\langle Tu^N\rangle| \le H_T\Gamma_\delta H_N$, $|\langle Yu^N\rangle| \le H_Y\Gamma_\delta H_N$, $|\langle \eta^L\rangle| \le H_L$. Let $w = (\langle \eta^L\rangle + \langle Tu^N\rangle,\ \eta^\omega + \langle Yu^N\rangle) \in \mathbb{C}^{n+1}$; in the max norm of $\mathbb{C}^{n+1}$, $|w| \le \max\{H_L + H_T\Gamma_\delta H_N, |\eta^\omega| + H_Y\Gamma_\delta H_N\}$.

Step S2.25: The bordered average system. *(new relative to* [K:bp:p2:triangular]*.)* Since $\det \mathcal{T} \neq 0$, $(\xi_0, \xi^\omega) := \mathcal{T}^{-1} w$ is the unique solution of

$$\langle T\rangle \xi_0 + \omega\, \xi^\omega = \langle \eta^L\rangle + \langle T u^N\rangle, \qquad \langle Y\rangle \xi_0 = \eta^\omega + \langle Y u^N\rangle, \tag{1}$$

and $|(\xi_0, \xi^\omega)| \leq |\mathcal{T}^{-1}||w| \leq S_*$ ([K:bp:p2:matrix](b)). This is (6.10) of HL19 in our sign convention (Remark 2.4). Note that $\langle T\rangle$ alone need not be invertible: the matrix $\mathcal{T}$ determines the constant ξ_0 and the frequency correction ξ^ω together.

Step S2.26: The energy equation and ξ^N. By Step S2.23, $\xi^N = \xi_0 - u^N$ is holomorphic, periodic on $T_{r-\delta}$ with $\partial_\omega \xi^N = -\eta^N$ and $|\xi^N| \leq |\xi_0| + \Gamma_\delta H_N \leq S_* + \Gamma_\delta H_N$. By linearity of the average and the second equation of (1), $\langle Y \xi^N\rangle = \langle Y\rangle \xi_0 - \langle Y u^N\rangle = \eta^\omega$. This is the third equation of (b) and the second one.

Step S2.27: The second cohomological equation. *(new relative to* [K:bp:p2:triangular]*.)* $g^L = T\xi^N + \omega\xi^\omega - \eta^L$ is holomorphic and periodic on $T_{r-\delta}$ with $|g^L| \leq H_T(S_* + \Gamma_\delta H_N) + |\omega| S_* + H_L$. Its average is, by linearity, $\langle T\rangle \xi_0 - \langle T u^N\rangle + \omega\xi^\omega - \langle \eta^L\rangle = 0$ by the first equation of (1). This is the point of the border: the frequency correction ξ^ω absorbs the part of the mean of $T\xi^N - \eta^L$ that the constant ξ_0 cannot adjust once it is spent on the energy equation. [K:bp:p2:cohomological](a) on $T_{r-\delta}$ with loss δ (valid since $\delta < r - \delta$) gives $\mathcal{R} g^L$ holomorphic, periodic on $T_{r-2\delta}$, with $\partial_\omega \mathcal{R} g^L = g^L$ and $|\mathcal{R} g^L| \leq \Gamma_\delta \sup|g^L|$. Then $T\xi^N - \partial_\omega \xi^L + \omega\xi^\omega = T\xi^N - g^L + \omega\xi^\omega = \eta^L$. With Steps S2.23 and S2.26 this is (a), (b), (c).

Step S2.28: Uniqueness. Let $(\tilde\xi^L, \tilde\xi^N, \tilde\xi^\omega)$ be as in (d). Then $\partial_\omega(\tilde\xi^N + u^N) = -\eta^N + \eta^N = 0$ on $T_{r-2\delta}$, and $\tilde\xi^N + u^N - \tilde c$, $\tilde c := \langle \tilde\xi^N + u^N\rangle$, is a zero-average holomorphic periodic solution of $Du\,\omega = 0$; by the uniqueness in [K:bp:p1:cohomological] (as in Step S2.21) it is 0. So $\tilde\xi^N = \tilde c - u^N$. Averaging the first equation of (b) for the tilde-solution ($\langle \partial_\omega \tilde\xi^L\rangle = 0$, [K:bp:p2:averages](b)) and using the third gives exactly (1) for $(\tilde c, \tilde\xi^\omega)$, whence $(\tilde c, \tilde\xi^\omega) = (\xi_0, \xi^\omega)$ by invertibility of $\mathcal{T}$, and $\tilde\xi^N = \xi^N$. Then $\partial_\omega(\tilde\xi^L - \xi^L) = 0$, so $\tilde\xi^L - \xi^L$ is constant, by the same uniqueness lemma applied to $\tilde\xi^L - \xi^L - \langle \tilde\xi^L - \xi^L\rangle$.

Step S2.29: Realness. If the data are real on reals, $\langle T\rangle, \langle Y\rangle, \langle \eta^L\rangle, \langle T u^N\rangle, \langle Y u^N\rangle$ are real ([K:bp:p2:averages](a)), $\mathcal{T}$ is a real matrix with real inverse ([K:bp:p1:real-inverse]), so ξ_0, ξ^ω are real; $\mathcal{R}$ preserves realness ([K:bp:p2:cohomological](a)), so u^N, ξ^N, g^L, ξ^L are real on reals. □

Remark 2.13 (Differences from HL19's printed statement). (i) HL19's $\mathfrak{R}_\omega$ discards the average of its argument, so their (6.9) $\xi^L = \xi_0^L + \mathfrak{R}_\omega(\eta^L - T\xi^N)$ omits the constant vector $\omega\xi^\omega$, which has no effect on non-zero Fourier modes; in the operator $\mathcal{R}$ of [K:bp:p2:cohomological] (defined on zero-average data) the term is carried explicitly in g^L, which costs the extra summand $|\omega| S_*$ in the bound of $\sup|g^L|$. HL19's bound for ξ^L (p. 46, last line) is true but is not obtained from [K:bp:p2:cohomological] as stated; we do not need it, because $|\omega| \leq \sigma_\omega |\omega^*|$ is available in the application. (ii) HL19 have real data and $\xi_0^L \in \mathbb{R}^n$; our lemma holds for complex data and records realness separately. (iii) The estimates of HL19 are in max of the two $\mathbb{C}^{n+1}$ block components, as here. (iv) Uniqueness (d) is not stated in HL19.

Lemma 2.14 (The linearized isoenergetic equation in the frame)**.** *Assume 2.1 with* $\det G_K \neq 0$ *on* T_r *and* $\omega \in \mathbb{R}^n$ *(any). Let* $\xi = (\xi^L, \xi^N)$ *be holomorphic on* $T_{r'}$, $r' \leq r$, *and* $\xi^\omega \in \mathbb{C}$*; put* $\Delta K = P\xi$, $\Delta\omega = -\omega\xi^\omega$*. Then on* $T_{r'}$*:*

(a) $QL = \left(\begin{smallmatrix} I_n \\ -\Omega_K \end{smallmatrix}\right)$*, hence* $QL\omega\,\xi^\omega = \left(\begin{smallmatrix} \omega \\ -\Omega_K \omega \end{smallmatrix}\right)\xi^\omega$.

(b) $Q\big(A\,\Delta K - \partial_\omega \Delta K - DK\,\Delta\omega\big) + QE = \Lambda\xi - \partial_\omega \xi + \binom{\omega}{0}\xi^\omega - \eta + E_{\mathrm{lin}}$*, where*

$$E_{\mathrm{lin}} := E_{\mathrm{red}}\xi + \Omega_0 E_{\mathrm{sym}}\,\partial_\omega \xi - \begin{pmatrix} 0 \\ \Omega_K \omega \end{pmatrix}\xi^\omega.$$

Consequently $A\Delta K-\partial_\omega\Delta K+L\omega\xi^\omega=-E$ holds exactly when the bordered system of Lemma 2.12(b) holds up to E_{lin}, i.e. when $\Lambda\xi-\partial_\omega\xi+\binom{\omega}{0}\xi^\omega=\eta-E_{\mathrm{lin}}$; the data are η^L,η^N of [K:`bp:p2:objects`], *T the torsion, and $Y=DH(K)N$. In particular, with ξ the solution of Lemma 2.12 (whose first two equations give $\Lambda\xi-\partial_\omega\xi+\binom{\omega}{0}\xi^\omega=\eta$), the residual of the linearized equation is $Q^{-1}E_{\mathrm{lin}}$, and $|E_{\mathrm{lin}}|\le|E_{\mathrm{red}}||\xi|+|E_{\mathrm{sym}}||\partial_\omega\xi|+|\Omega_K||\omega||\xi^\omega|$, all quadratic in (E,ξ) by* [K:`bp:p2:geometry`].
Source: HL19 (6.4)–(6.5), p. 45, with $\Omega=\Omega_0$, $A=0$ (canonical Case III); (6.34), p. 49 for E_{lin}. Frame identities: [K:`bp:p2:frame-identities`].

Proof. **Step S2.30: The product QL.** $Q=-\Omega_0P^\top\Omega_0$ and $P^\top\Omega_0L=\binom{L^\top\Omega_0L}{N^\top\Omega_0L}=\binom{\Omega_K}{I_n}$ by [K:`bp:p2:frame-identities`](a). By [K:`bp:p2:matrix`](d), $\Omega_0\binom{U_1}{U_2}=\binom{-U_2}{U_1}$, so $QL=-\binom{-I_n}{\Omega_K}=\binom{I_n}{-\Omega_K}$. This is (a).

Step S2.31: Expanding the linearized operator. $\partial_\omega(P\xi)=(\partial_\omega P)\xi+P\partial_\omega\xi$, so $A\Delta K-\partial_\omega\Delta K=(AP-\partial_\omega P)\xi-P\partial_\omega\xi$, and $-DK\,\Delta\omega=L\omega\,\xi^\omega$. By [K:`bp:p2:frame-identities`](e), $Q(AP-\partial_\omega P)=\Lambda+E_{\mathrm{red}}$; by (f), $QP=I_{2n}-\Omega_0E_{\mathrm{sym}}$; by (a), $QL\omega\xi^\omega=\binom{\omega}{0}\xi^\omega-\binom{0}{\Omega_K\omega}\xi^\omega$. Adding $QE=-\eta$: $Q(\dots)+QE=(\Lambda+E_{\mathrm{red}})\xi-\partial_\omega\xi+\Omega_0E_{\mathrm{sym}}\partial_\omega\xi+\binom{\omega}{0}\xi^\omega-\binom{0}{\Omega_K\omega}\xi^\omega-\eta$, which is (b).

Step S2.32: Consequence. If the border system holds, (b) gives $Q\,\mathrm{Res}=E_{\mathrm{lin}}$ for the residual $\mathrm{Res}:=A\Delta K-\partial_\omega\Delta K+L\omega\xi^\omega+E$. $Q=-\Omega_0P^\top\Omega_0$ is invertible iff P is, and $\det(P)^2\det\Omega_0=\det(P^\top\Omega_0P)=\det(\Omega_0+E_{\mathrm{sym}})\neq0$ when $|E_{\mathrm{sym}}|<1$ (Neumann series [K:`bp:p1:neumann`] applied to $I-\Omega_0E_{\mathrm{sym}}$, $|\Omega_0E_{\mathrm{sym}}|=|E_{\mathrm{sym}}|$, [K:`bp:p1:matrix-norms`]). The bound on E_{lin} is [K:`bp:p2:matrix`](b), (c). □

2.5. **Suspected errata and sign checks.**

Remark 2.15 (Candidate errata of HL19, checked against the pinned PDF (`arxiv_1711.05515.pdf`, v1)). All three were checked on the rendered pages 18, 19, 46 (not only on the text layer).

E1 (strict inequalities).: Thm 3.6, (3.16), p. 19: "$\|K_\infty-K\|_{\rho_\infty}<\dfrac{\mathfrak{C}_2\|E_c\|_\rho}{\gamma^2\rho^{2\tau}}$, $|\omega_\infty-\omega|<\dfrac{\mathfrak{C}_3\|E_c\|_\rho}{\gamma^2\rho^{2\tau}}$"; likewise (3.11), p. 18 for Thm 3.1. With $E_c\equiv0$ the right side is 0, and (3.15) holds, but no K_∞ satisfies $\|K_\infty-K\|<0$. The statements must read "$\le$" (the frozen I-G1 does).

E2 (sign in (3.12)).: p. 18: "(3.12) $X_h(K(\theta))=\mathfrak{L}_\omega K(\theta)$, $\langle c\circ K\rangle=c_0$". With $\mathfrak{L}_\omega=-\partial_\omega$ (4.1), p. 20, this reads $X_h(K)=-DK\omega$, contradicting the invariance equation (2.1), p. 6, $X_h(K)=DK\,\omega$, and the error (2.2), p. 7, $E=X_h(K)-DK\omega$. The correct form is $X_h(K)+\mathfrak{L}_\omega K=0$, which is how p. 45 (the total error before (6.1)) and p. 48 (Lemma 6.2, $\bar E_c$) write it. Our Remark 2.4 uses the consistent convention.

E3 (Rem. 3.7, first equality).: p. 19: "$Dh(K(\theta))N(\theta)=X_h(K(\theta))^\top\Omega(K(\theta))N(\theta)=-\omega^\top DK(\theta)^\top\Omega(K(\theta))N(\theta)=(\omega^\top\ 0_{n-d}^\top)=\hat\omega^\top$". By Rem. 2.1, p. 6, $\Omega X_h=(Dh)^\top$, so $Dh=(\Omega X_h)^\top=-X_h^\top\Omega$: the first equality misses a minus sign. The second equality is likewise wrong for $X_h=+DK\omega$ (it would need $X_h=-DK\omega$, the E2 convention). The two slips cancel and the final value $Dh\,N=\hat\omega^\top$ is correct, as Lemma 2.8(a) shows (using $L^\top\Omega_0N=-I_n$, [K:`bp:p2:frame-identities`](a)); an independent check at the trivial torus is Lemma 2.6, Step S2.9. The correct chain is $Dh\,N=-X_h^\top\Omega N=-\omega^\top DK^\top\Omega N=\omega^\top$.

Other points checked and found consistent (no erratum): (6.7)–(6.10), p. 46, with $\mathfrak{L}_\omega=-\partial_\omega$, $\mathfrak{R}_\omega=-\mathcal{R}$ (Remark 2.4); (6.3) and $\bar\omega=(1-\xi^\omega)\omega$ (p. 50, (6.32)); the equation $QL\omega=\binom{\omega}{0}-\binom{0}{\Omega_K\omega}$ against the display on p. 46 (Lemma 2.14(a)); the sign $\Delta\omega=-\omega\xi^\omega$. HL19's (6.9) omits the constant $\omega\xi^\omega$ (Remark 2.13(i)), which is harmless.

Remark 2.16 (Doubts about the frozen statements). None of the frozen statements used in this part looks false. Two reading points. (a) The frozen B_h uses the real derivative of $\operatorname{Re} h$ via `frequencyMap`; Lemma 2.6 identifies it with the complex Hessian of h only under "h real on the real points of a neighbourhood of I", which I-HEAD assumes on all of B_{r_0}; without that hypothesis T_h and B_h can differ by the imaginary part. (b) The frozen last row of T_H is $DH(K)N$, not $\omega^\top$ (Definition 1.6); Lemma 2.8 shows the difference is $(\eta^L)^\top = O(e)$, so the two readings give the same I-G1 up to the choice of σ_T.